1) a)
Shear, with the x-axis fixed, with $(0,1)$ mapped to $(2,1)$

1) b)
Shear, with the y-axis fixed, with $(1,0)$ mapped to $(1,3)$

1) c)
Shear, with the x-axis fixed, with $(0,1)$ mapped to $(-4,1)$

1) d)
Shear, with the y-axis fixed, with $(1,0)$ mapped to $\left(1,-\frac{3}{2}\right)$

2)
This is a Shear, with the x-axis fixed, with $(0,1)$ mapped to $(6,1)$, so area is preserved.
Hence the area of the shape's image is 4 units^2

3)
$\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}\begin{pmatrix}-2\\ 4\end{pmatrix}=\begin{pmatrix}18\\ 4\end{pmatrix}\Rightarrow\begin{pmatrix}-2+4k\\ 4\end{pmatrix}=\begin{pmatrix}18\\ 4\end{pmatrix}$
$-2+4k=18\Rightarrow k=5$

4)
$\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}\begin{pmatrix}22\\ -6\end{pmatrix}=\begin{pmatrix}22\\ -94\end{pmatrix}\Rightarrow\begin{pmatrix}22\\ 22k-6\end{pmatrix}=\begin{pmatrix}22\\ -94\end{pmatrix}$
$22k-6=-94\Rightarrow k=-4$

5)
Triangle X has area $\frac{1}{2}\times 7\sqrt{2}\times\sqrt{2}=7$ units^2
$\begin{pmatrix}1 & 8\sqrt{2}\\ 0 & 1\end{pmatrix}\begin{pmatrix}-1 & -3\sqrt{2} & 4\sqrt{2}\\ \sqrt{2} & 0 & 0\end{pmatrix}=\begin{pmatrix}15 & -3\sqrt{2} & 4\sqrt{2}\\ \sqrt{2} & 0 & 0\end{pmatrix}$
So X' has vertices at $\left(15,\sqrt{2}\right)$, $\left(-3\sqrt{2},0\right)$ and $\left(4\sqrt{2},0\right)$
Triangle X' has area $\frac{1}{2}\times 7\sqrt{2}\times\sqrt{2}=7$ units^2

6)

$\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}\begin{pmatrix}8\\ 3\sqrt{3}\end{pmatrix}=\begin{pmatrix}8\\ 5\sqrt{3}\end{pmatrix}\Rightarrow\begin{pmatrix}8\\ 8k+3\sqrt{3}\end{pmatrix}=\begin{pmatrix}8\\ 5\sqrt{3}\end{pmatrix}$

$8k+3\sqrt{3}=5\sqrt{3}\Rightarrow 8k=2\sqrt{3}\Rightarrow k=\frac{\sqrt{3}}{4}$

$\therefore \mathbf{A}=\begin{pmatrix}1 & 0\\ \frac{\sqrt{3}}{4} & 1\end{pmatrix}$

7)

$\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}\begin{pmatrix}50\\ 4\end{pmatrix}=\begin{pmatrix}4\\ 4\end{pmatrix}\Rightarrow\begin{pmatrix}50+4k\\ 4\end{pmatrix}=\begin{pmatrix}4\\ 4\end{pmatrix}$

$50+4k=4\Rightarrow 4k=-46\Rightarrow k=-\frac{23}{2}$

$\therefore \mathbf{A}=\begin{pmatrix}1 & -\frac{23}{2}\\ 0 & 1\end{pmatrix}$

8)

$\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}\begin{pmatrix}45\\ 9\end{pmatrix}=\begin{pmatrix}0\\ p\end{pmatrix}\Rightarrow\begin{pmatrix}45+9k\\ 9\end{pmatrix}=\begin{pmatrix}0\\ p\end{pmatrix}$

$45+9k=0\Rightarrow 9k=-45\Rightarrow k=-5$

$p=9$

$\therefore \mathbf{A}=\begin{pmatrix}1 & -5\\ 0 & 1\end{pmatrix}$

9)

As $(5,0)$ is mapped to itself, the shear must fix the x-axis.

$\begin{pmatrix}1 & k\\ 0 & 1\end{pmatrix}\begin{pmatrix}3\\ 20\end{pmatrix}=\begin{pmatrix}11\\ 20\end{pmatrix}\Rightarrow\begin{pmatrix}3+20k\\ 20\end{pmatrix}=\begin{pmatrix}11\\ 20\end{pmatrix}$

$3+20k=11\Rightarrow 20k=8\Rightarrow k=\frac{2}{5}$

$\therefore \mathbf{A}=\begin{pmatrix}1 & \frac{2}{5}\\ 0 & 1\end{pmatrix}$

10) a)

As $(-12,8)$ is mapped to $(-12,-4)$, the shear must fix the y-axis.

$\begin{pmatrix}1 & 0\\ k & 1\end{pmatrix}\begin{pmatrix}-12\\ 8\end{pmatrix}=\begin{pmatrix}-12\\ 17\end{pmatrix}\Rightarrow\begin{pmatrix}-12\\ -12k+8\end{pmatrix}=\begin{pmatrix}-12\\ 17\end{pmatrix}$

$-12k+8=17\Rightarrow -12k=9\Rightarrow k=-\frac{3}{4}$

$\therefore \mathbf{A}=\begin{pmatrix}1 & 0\\ -\frac{3}{4} & 1\end{pmatrix}$

10) b)

As $(-12, 17)$ is mapped to $(-12, 8)$, the shear must fix the y-axis.

$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} -12 \\ 17 \end{pmatrix} = \begin{pmatrix} -12 \\ 8 \end{pmatrix} \Rightarrow \begin{pmatrix} -12 \\ -12k + 17 \end{pmatrix} = \begin{pmatrix} -12 \\ 8 \end{pmatrix}$

$-12k + 17 = 8 \Rightarrow -12k = -9 \Rightarrow k = \frac{3}{4}$

$\therefore \mathbf{B} = \begin{pmatrix} 1 & 0 \\ \frac{3}{4} & 1 \end{pmatrix}$

10) c)

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ -\frac{3}{4} & 1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ \frac{3}{4} & 1 \end{pmatrix} = \begin{pmatrix} 1 \times 1 + 0 \times \frac{3}{4} & 1 \times 0 + 0 \times 1 \\ -\frac{3}{4} \times 1 + 1 \times \frac{3}{4} & -\frac{3}{4} \times 0 + 1 \times 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \mathbf{I}$$

11)

The transformation is a shear, and clearly it is with the y-axis fixed, with $(1, 0)$ mapped to $(1, k)$

$\begin{pmatrix} 1 & 0 \\ k & 1 \end{pmatrix}\begin{pmatrix} 2 & 4 & 7 & 5 \\ 2 & 6 & 6 & 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 8.25 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2 & 4 & 7 & 5 \\ 2k + 2 & 4k + 6 & 7k + 6 & 5k + 2 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 8.25 \end{pmatrix}$

$\therefore a = 2, b = 4, c = 7$

$5k + 2 = 8.25 \Rightarrow k = 1.25$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & 0 \\ 1.25 & 1 \end{pmatrix}$

$A'(2, 4.5)$, $B'(4, 11)$, $C'(7, 14.75)$

12)

The transformation is a shear, and clearly it is with the x-axis fixed, with $(0, 1)$ mapped to $(k, 1)$

$\begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}\begin{pmatrix} -1 & -5 & -5 & -1 \\ 3 & 6 & 9 & 9 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 9 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -1 + 3k & -5 + 6k & -5 + 9k & -1 + 9k \\ 3 & 6 & 9 & 9 \end{pmatrix} = \begin{pmatrix} a & b & c & 5 \\ d & e & f & 9 \end{pmatrix}$

$\therefore d = 3, e = 6, f = 9$

$-1 + 9k = 5 \Rightarrow k = \frac{2}{3}$

$\therefore \mathbf{M} = \begin{pmatrix} 1 & \frac{2}{3} \\ 0 & 1 \end{pmatrix}$

$A'(1, 3)$, $B'(-1, 6)$, $C'(1, 9)$

13) a)
Shear, with the y-axis fixed, with $(1, 0)$ mapped to $(1, k)$

13) b)
$\det \mathbf{A} = 1 \times 1 - 0 \times k = 1$

13) c)
As $\det \mathbf{A} = 1$, both area and orientation are preserved by the transformation.